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Apparatus and method for direct printing on corrugated metal clad cable

Abstract

An apparatus and method for direct printing on corrugated metal clad cable. The apparatus includes an entrance having one or more entrance rollers, a set of lower rollers, a series of printheads near the entrance, a UV curing station downstream of the printheads, one or more exit rollers, a print controller and a series of ink tanks. The entrance and exit rollers maintain the cable in proper position and alignment for the printheads. The method for printing provides for clear and legible printing on the cable.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10493777	12/2018	McConville	N/A	B41J 11/00216
2010/0101716	12/2009	Roof	156/247	B41J 11/00214

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION DATA (1) This application claims the benefit of and priority to Provisional U.S. Patent application Ser. No. 63/302,312, filed Jan. 24, 2022, titled, APPARATUS AND METHOD FOR DIRECT PRINTING ON CORRUGATED METAL CLAD CABLE, and Provisional U.S. Patent application Ser. No. 63/319,518, filed Mar. 14, 2022, titled, APPARATUS AND METHOD FOR DIRECT PRINTING ON CORRUGATED METAL CLAD CABLE, the disclosures of which are incorporated herein in their entireties.

BACKGROUND

(1) The present disclosure relates generally to an apparatus and method for printing on metal clad cable, and more particularly, to an apparatus and method and for direct fixed and variable printing on corrugated metal clad cable.

(2) Metal clad cable has a spiral wound, interlocked or corrugated metal cladding or armor that may provide a low impedance ground path or grounding conductor for equipment. The cladding or armor is spiral wound with edges interlocking adjacent edges that also provides continuous protection for the wires within the cladding. Typical metal clad (MC) cables can include power conductors, an insulated grounding conductor, bare grounding/bonding wire(s), signal or control conductors and the like.

(3) It is often necessary to provide marking on the cable to indicate the type(s) of conductors, conductor sizes, whether the cable includes a tape or tapes, and a length markings along the cable. Markings may also be required to meet certain industry codes or standards.

(4) There are various known ways in which markings are provided on cables. For example, some cables include tapes affixed to the surface of the cable armor. Other ways in which cables are marked include application of paint or printing directly on the cable armor, etching or embossing the armor.

(5) There are, however, drawbacks to these known systems. For example, physical tapes affixed to the armor can inadvertently fall off of the armor. This is particularly the case when the cables are

manipulated, as when they are pulled through conduit or wall openings. In addition, the physical tapes can require an adhesive or an additional adhesive tape to secure the tape to the cable armor. (6) Painting on the cable armor, while it may provide permanent marking, may not dry properly during the cable fabrication process and as such the paint may smudge or smear, making the marking unreadable or unreliable. Etching and embossing require specialized equipment which can add costs and time, as well as equipment maintenance.

(7) Accordingly, it is desirable to provide a method and apparatus for printing on metal clad cable. Desirably, such a method and apparatus provide printing that meets required industry codes and standards. More desirably still, such a method and apparatus provide for printing direct fixed and variable markings on metal clad cable, which markings are clear and do not smudge or smear upon manufacture or use.

SUMMARY

(8) According to one aspect an apparatus for direct printing on corrugated metal clad cable includes an entrance having one or more entrance rollers, a set of lower rollers, a series of printheads near the entrance, a UV curing station downstream of a last of the series of printheads, one or more exit rollers, a print controller and a series of ink tanks.

(9) The apparatus provides clear and legible printing on the cable. The printing can include indicia or markings, such as a distance or length marking, a color code indicating the clad material, markings that indicate the number and color of the conductors and the direction of pull of the cable, indicators showing the conductor gauge and number of conductors, and the like.

(10) In embodiments of the apparatus, the lower rollers can be V-groove rollers and can include a low friction surface. The apparatus can include a gap between the last printhead of the series of printheads and the UV curing station.

(11) The entrance rollers can be V-groove rollers. The entrance and/or exit rollers can be biased downwardly toward the cable. The rollers can be mounted to arms that are biased, as by springs toward the cable.

(12) The apparatus can include multiple print controllers.

(13) In another aspect, a method for direct printing on corrugated metal clad cable includes providing a printing system having one or more entrance rollers, a series of lower rollers, a series of printheads, a UV curing station, one or more exit rollers and one or more print controllers. The method includes transporting the cable into the printing system, passing the cable through the entrance rollers and along the lower rollers, moving the cable past the printheads, and printing one or more markings on the cable, moving the cable through a space between a last of the series of printheads and the UV curing station, moving the cable through the UV curing station to cure the one or more markings on the cable and passing the cable through the exit rollers.

(14) In methods, the entrance, exit and/or lower rollers can be V-groove rollers. The entrance and exit rollers can be biased downwardly toward the cable. The entrance and exit rollers can be mounted to arms and the arms can be biased, as by springs, toward the cable. The method can include biasing the cable toward the lower rollers. Other types of rollers can be used to maintain the cable properly distanced from and aligned with the printheads.

(15) In methods, the markings can include one or more of a distance or length marking, a clad material marking, a conductor number and color marking, a direction of pull of the cable marking, and a conductor gauge and material marking. The clad material marking and/or the conductor number and color marking can be color coded markings. The number and color of the conductors marking and the direction of pull of the cable marking can be the same marking. The conductor gauge and material marking can be the same marking. The number and color of the conductors marking and the direction of pull of the cable marking can be, for example, chevrons.

(16) These and other features and advantages of the present invention will be apparent from the following detailed description, in conjunction with the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an illustration of an example of a metal clad cable having printing thereon according to an embodiment described herein;
- (2) FIG. 2 is a rear view of a system for printing on metal clad cable as viewed looking upstream of the system;
- (3) FIG. 3 is a perspective illustration an example of an apparatus for fabricating metal clad cable;
- (4) FIG. 4 is a front view of the printing system as viewed looking upstream of the system;
- (5) FIG. 5 is a perspective view of the rear of the printing system as viewed looking downstream of the system;
- (6) FIG. 6 is a view of the cable take-up portion of the cable fabrication system;
- (7) FIG. 7 is another front view of the printing system as viewed looking downstream of the system;
- (8) FIG. 8 is a front view of the printing system;
- (9) FIG. 9 is a front view of the printing system illustrating an embodiment of an entrance roller and a lower roller;
- (10) FIG. 10 is perspective rear view of the entrance roller of FIG. 9;
- (11) FIG. 11 is view along the cable looking downstream into the printing system;
- (12) FIG. 12 is view along the top side of the cable looking upstream into the printing system;
- (13) FIG. 13 is view along the bottom side of the cable looking upstream into the printing system;
- (14) FIG. 14 is a front view of the printing system illustrating an embodiment of an exit roller and a lower roller;
- (15) FIG. 15 is a rear view of the printing system;
- (16) FIG. 16 is an enlarged lower rear view of the printing system; and
- (17) FIG. 17 is a schematic illustration of a storage, feed, print and take-up arrangement for direct printed corrugated metal clad cable.

DETAILED DESCRIPTION

(18) While the present device is susceptible of embodiment in various forms, there is shown in the figures and will hereinafter be described a presently preferred embodiment with the understanding that the present disclosure is to be considered an exemplification of the device and is not intended to be limited to the specific embodiment illustrated.

(19) Referring to the figures, and in particular to FIG. 1, there is shown an example of a metal clad or armored corrugated cable **10** having markings or indicia **12** thereon. The markings **12** can include, for example, a distance or length marking **14**, a color code **16** indicating the clad material, which in the illustrated embodiment is a blue marking indicating a steel clad, a series of markings, which can be, for example as illustrated, in the form of chevrons **18**, that indicate the number and color of the conductors **C** and the direction of pull of the cable **10**, and numeric indicators **20** showing the conductor **C** gauge and number of conductors **C**. The color of the numeric indicators **20** indicates the armor material.

(20) All of the markings **12** are made directly on the metal clad **22** using an ultraviolet (UV) curable ink formulation. The distance or length marking **14** is variable with the length of the cable **10** in a particular or given run of cable **10**. That is the distance or length marking **14** will change along the length of the cable **10** being marked. It is anticipated that only one particular type of cable **10** will be fabricated at a time. As such, while the distance or length marking **14** is variable, it is anticipated that the other markings, including, for example, the color **16** code indicating the clad material and a measuring (distance or length) mark, the series of markings (the illustrated chevrons **18**), that indicate the number and color of the conductors **C** and the direction of pull of the cable **10**, and the numeric indicators **20** showing the conductor **C** gauge and number of conductors **C** may all

be fixed markings or indicia for a given cable **10** fabrication run.

(21) FIGS. **2** and **3** illustrate a portion of the cable fabrication **30** and printing **40** systems. The fabrication system **30** is well known in the art. Briefly, conductors **C** are drawn into the fabrication system **30** in which the clad or armor **22** is wound around the conductors **C** and as desired, tape **T**, bare grounding/bonding wire(s) **G**, signal or control conductors and the like, as desired for the particular cable **10** configuration.

(22) As the cable **10** exits the cable fabrication system **30**, it is wound around a drum **32** to maintain a desired tension and fabrication rate. As seen in FIGS. **4** and **5**, in the illustrated process, the now formed cable **10** enters the printing system **40**. The entrance **44** of the printing system **40** includes one or more entrance rollers **46** to maintain the cable **10** in proper horizontal placement, e.g., a horizontal plane centered, in the printing system **40**. The entrance roller **46** can be mounted to bias the cable **10** downwardly. In a current embodiment, the entrance roller **46** is mounted to an arm **48** and the arm **48** is biased, as by a spring (not shown) downwardly. The downward bias of the entrance roller **46**, in conjunction with a first lower roller **64a** maintains the cable **10** in proper vertical placement, e.g., vertical plane, to maintain the cable **10** a proper distance from the printheads **52**. In a current embodiment, the entrance and first lower roller **46**, **64a** (as well as the remaining lower rollers **64**) can be V-groove rollers to provide both centering and hold-down functions.

(23) Following the entrance roller **46**, the system **40** includes a series of printheads **52** (see FIGS. **7-10**) near the entrance **44** to the printing system **40**. The printheads **52** each print a specific, individual color of ink onto the cable **10** as the cable **10** moves past the printhead **52**. In a current embodiment, the colors include a blue ink, a black ink, a white ink, a red ink and a green ink. It will be appreciated that any ink colors can be used in the system **40** and method.

(24) The printing system **40** further includes a series of ink tanks **58** corresponding to the ink colors used and a series of controllers **60** and pumps **62** for controlling the application of the inks. The system **40** may be configured such a controller **60** controls the application of multiple inks.

(25) The system **40** also includes a series of lower rollers **64** (as briefly discussed above), along which the cable **10** moves during the printing process. The lower rollers **64** may have a low friction surface **66**, such as a Teflon surface. The lower rollers **64** can, like the entrance roller **46**, be V-groove rollers to maintain horizontal alignment of the cable **10** and, with the entrance roller **46**, maintain proper vertical alignment of the cable **10** relative to the printheads **52** during the printing process. In a current embodiment, the lower rollers **64** are idler rollers and do not provide a drive force for the cable. One or more of the lower rollers **64** may also be a driven roller.

(26) A UV curing station **70** is located downstream of the printheads **52**. In order to provide a measure of natural curing, the UV station **70** is located a distance downstream of the printheads **52**; that is, there is a gap **72** between the last pinhead **52e** and the UV station **70**. A final lower roller **64d** is located downstream of the curing station **70**. In a current embodiment the printing system **40** includes one or more exit rollers **74** downstream of the final lower roller **64d**.

(27) The exit roller **74** can, like the entrance roller **46** and lower rollers **64**, be a V-groove roller. The exit roller **74** maintains the cable **10** in proper horizontal placement, e.g., centered, in the printing system **40** relative to the printheads **52**, and along with the final lower roller **64d**, maintains the cable **10** in proper vertical placement, e.g., maintains the cable **10** a proper distance from the printheads **52**. The exit roller **74** can, like the entrance roller **46**, be mounted to an arm **76** and the arm **76** can be biased, as by a spring (not shown), to maintain the cable **10** in proper vertical placement, e.g., vertical plane, to maintain the cable **10** a proper distance from the printheads **52**. The V-groove roller **74** provides both centering and hold-down functions.

(28) As illustrated, the entrance and exit rollers **46**, **74** and the lower rollers **64** help to maintain the cable in proper horizontal and vertical planes to maintain the cable **10** properly distanced and aligned with the printheads **52**. It has also been found that excessive vibration can be induced in the cable **10** and that the entrance and exit rollers **46**, **74** dampen the vibration to facilitate maintaining

the cable the still (e.g., reduce vibration) as well as properly distanced relative to, and aligned with, the printheads **52** to facilitate clear and legible printing on the cable **10**.

(29) It has also been found that providing a gap **72** between the last printhead **52e** and the UV curing station **70** also helps setting of the ink prior to UV curing, so that when the cable **10** exits the printing system **40** the ink is fully cured and does not smear or smudge when it is subsequently stored on a spool **80**. It has also been found that because the cable **10**, as it is wound onto the spool **80**, is moved laterally across the spool **80**, the rollers maintain the cable **10** in a proper horizontal position relative to the printheads **52**.

(30) The printing system **40** is a modular system. As such, the entrance and exit rollers **46**, **74**, lower rollers **64**, printheads **52**, ink tanks **58**, controllers **60** and pumps **62** can be readily removed and replaced with minimal impact on the overall cable fabrication system and production timing.

(31) It is anticipated that the printing system **40** can be separate and apart from the cable fabrication system **30**. That is, the printing system **40** does not have to be immediately downstream of the cable fabrication system **40**. Indeed, in an anticipated arrangement, for example, following fabrication, the cable **10** can be stored on a spool, such as spool **82**. The spool **82** can then at, for example a later time or different location, be used as the feed to the printing system **40**.

(32) An example of such a system is shown schematically in FIG. **17**. The cable **10** is fed from the spool **82** into the printing system **40** at entrance roller **46**. A cable tensioner **84** can be positioned upstream of the printing system **40** to provide sufficient tension to the cable **10** as it is fed into the printing system **40**. It is envisioned that the feed or payoff spool **82** will be an idler spool and as such, a cable tensioner **84** between the payoff spool **82** and the printing system **40** would be desirable to maintain proper tension in the cable as **10** it approaches the entrance roller **46** and is fed into the printing system **40**. In an anticipated system, a biased dancer, such as a spring actuated pivoting or reciprocating dancer arm, or arms, can be used to provide the desired tension in the cable **10** without over-tensioning the cable **10** as it is fed into the printing system **40**. The cable **10** can then be taken up on at take-up spool or reel **86** for subsequent processing.

(33) The ink used for the printing is a UV curable ink that can be pigmented or unpigmented. A suitable ink is a LED UV curable ink can be pigmented or unpigmented, such as that commercially available from Squid Ink Manufacturing of Spring Lake, MN under product numbers SI-PZ8000-500 (black ink), SI-PZ8001-500 (red ink), SI-PZ8003-500 (green ink), SI-PZ8010-500R (white ink) and SI-PZ81620ZDEV-500R (blue/turquoise ink). Other colors of ink can of course be used, as can other types/manufacturers inks.

(34) A method for printing on a metal clad or metal armored corrugated cable **10** includes providing a printing system **40** having one or more entrance rollers **46**, a series of lower rollers **64**, a series of printheads **52**, a UV curing station **70**, one or more exit rollers **74**, ink tanks **58**, and one or more print controllers **60**. The cable **10** is driven through the printing system **40** through the entrance roller **46**, along the lower rollers **64**.

(35) The method includes maintaining the cable **10** in a desired horizontal position and a desired vertical position relative to the printheads **52** and moving the cable **10** past the printheads **52**, and printing one or more markings **12** on the cable **10**. Subsequent to printing, the cable **10** is moved through the gap **72** and then through the UV curing station **70**. Following UV curing, the cable **10** passes through the exit roller **74**.

(36) In methods, the indicia **12** includes one or more of a distance or length marking **14**, a marking **16** indicating the clad material, a marking **18** indicating the number and color of the conductors and a direction of pull of the cable, and a marking **20** indicating the conductor gauge.

(37) In methods, the marking **16** indicating the clad material and the marking **18** indicating the number and color of the conductors is a color coded marking. In methods, the marking **18** indicating the number and color of the conductors and the marking indicating the direction of pull of the cable are the same marking.

(38) It will be understood that the features from any one of the embodiments described above may

be implemented in, combined or used together with, or replace features from any of the other embodiments described above. Additionally, although V-groove rollers are described, other types of rollers will be recognized by those skilled in the art and can be used to maintain the cable properly distanced from and aligned with the printheads.

(39) All patents referred to herein, are hereby incorporated herein by reference, whether or not specifically done so within the text of this disclosure.

(40) In the present disclosure, the words “a” or “an” are to be taken to include both the singular and the plural. Conversely, any reference to plural items shall, where appropriate, include the singular. In addition, it is understood that terminology referring to orientation of various components, such as “upper” or “lower” is used for the purposes of example only, and does not limit the subject matter of the present disclosure to a particular orientation.

(41) From the foregoing it will be observed that numerous modifications and variations can be effectuated without departing from the true spirit and scope of the novel concepts of the present disclosure. It is to be understood that no limitation with respect to the specific embodiments illustrated is intended or should be inferred. The disclosure is intended to cover all such modifications as fall within the scope of the claims.

Claims

1. A method for direct printing on corrugated metal clad cable comprising: providing a printing system having one or more entrance rollers, a series of lower rollers, a series of printheads, a UV curing station, one or more exit rollers and one or more print controllers; transporting the cable into the printing system, passing the cable through the entrance rollers and along the lower rollers; moving the cable past the printheads, and printing one or more markings on the cable; moving the cable through a space between a last of the series of printheads and the UV curing station; moving the cable through the UV curing station to cure the one or more markings on the cable; and passing the cable through the exit rollers.
 2. The method of claim 1, wherein the rollers of the set of lower rollers include a low friction surface.
 3. The method of claim 1, wherein the rollers of the set of lower rollers are V-groove rollers.
 4. The method of claim 1, wherein the one or more entrance rollers are V-groove rollers.
 5. The method of claim 1, wherein the one or more exit rollers are V-groove rollers.
 6. The method of claim 5, wherein the one or more exit rollers are V-groove rollers.
 7. The method of claim 1, wherein the markings include one or more of a distance or length marking, a clad material marking, a conductor number and color marking, a direction of pull of the cable marking, and a conductor gauge marking.
 8. The method of claim 7, wherein the clad material marking and/or the conductor number and color marking are color coded markings.
 9. The method of claim 7, wherein the number and color of the conductors marking and the direction of pull of the cable marking are the same marking.
 10. The method of claim 7, wherein the number and color of the conductors marking and the direction of pull of the cable marking are chevrons.
 11. The method of claim 1, including biasing the cable toward the lower rollers.
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